Ders 13 — Açı modülasyonunun lineersizliğe dayanıklılığı

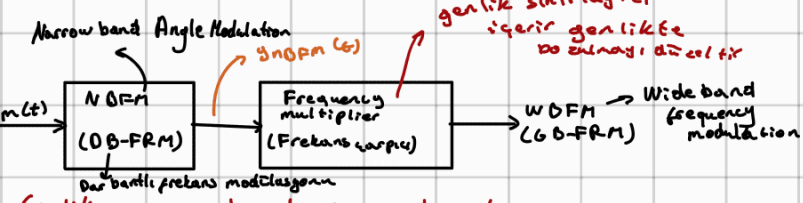
$$\cos^3(\alpha) = \frac{3}{4}\cos(\alpha) + \frac{1}{4}\cos(3\alpha)$$

$|\phi(t)| \ll 1 \Rightarrow \cos\phi(t) \cong 1 \quad \sin\phi(t) \cong \phi(t)$

Create FM

1) Dolaylı metod (Indirect method)

(Armstrong tekniği)



- Genlik ve faz bozulmasına sebep olur

$$y(t) = A_c \cos(2\pi f_c t + \phi(t))$$

$$= A_c\left[\cos(2\pi f_c t)\cos(\phi(t)) - \sin(2\pi f_c t)\sin(\phi(t))\right]$$

$\cong 1$ $\quad$ $\cong \phi(t)$

$$y(t) = A_c\left[\cos(2\pi f_c t) - \phi(t)\sin(2\pi f_c t)\right]$$

NBFM when $\phi(t) = 2\pi k_f \int m(z)\,dz$

$$a\cos x + b\sin x = c\cos(x+\theta)$$
$$c = \sqrt{a^2+b^2},\quad \theta = \tan^{-1}(-b/a)$$

$$y_{NBFM}(t) = A_c E(t)\cos(2\pi f_c t + \alpha(t))$$

$$E(t) = \sqrt{1+\phi^2(t)},\quad \alpha(t) = \arctan(\phi(t))$$

$$\phi(t) = 2\pi k_f \int m(z)\,dz$$

$$f_i(t) - f_c = \frac{1}{2\pi}\frac{d\alpha(t)}{dt} = \frac{1}{2\pi}\frac{\dot\phi(t)}{1+\phi^2(t)} = \frac{1}{2\pi}\frac{2\pi k_f m(t)}{1+\phi^2(t)}$$

$$f_i(t) - f_c = k_f m(t)\underbrace{\left[1-\phi^2(t)+\phi^4(t)-\right]}_{1/(1+\phi^2(t))}$$

2) Direct Methode

Voltage controlled oscillator

Hartley ve Colpitts osilatörlerinde osilasyon frekansı

Gko

$$f_i(t) = \frac{1}{2\pi\sqrt{LC}},\quad C = C_0 - k\,m(t)$$

varactor

$$= \frac{1}{2\pi\sqrt{L[C_0 - k m(t)]}} = \frac{1}{2\pi\sqrt{LC_0\left[1-\frac{k m(t)}{C_0}\right]}}$$

$$= \frac{1}{2\pi\sqrt{LC_0}}\cdot\frac{1}{\left[1-\frac{k m(t)}{C_0}\right]^{1/2}} = f_c\left[1-\frac{k m(t)}{C_0}\right]^{-1/2}$$

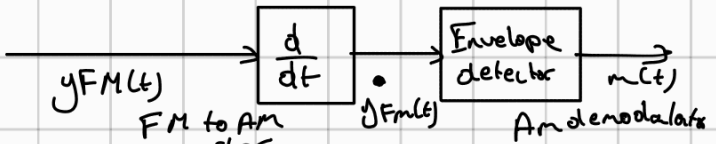
f_c * $|x| \ll 1 \Rightarrow (1+x)^n \approx 1+nx$

$$\frac{k m(t)}{C_0} \ll 1 \rightarrow f_i(t) \cong f_c\left[1+\frac{k m(t)}{2C_0}\right]$$

$$f_i(t) = f_c + \frac{k f_c}{2C_0} m(t)$$

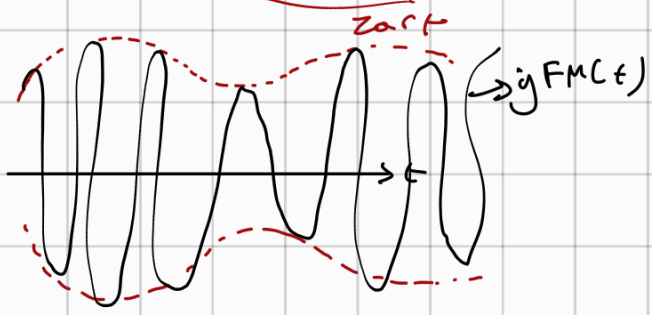
* frekans kararlılığı çok iyidir

FRM (FM)'nin Dönüşümü



$$y_{FM}(t) = A_c\cos\left(2\pi f_c t + 2\pi k_f \int_{-\infty}^{t} m(z)\,dz\right)$$

$$\frac{dy_{FM}(t)}{dt} = -A_c\left[2\pi f_c + 2\pi k_f m(t)\right]\sin(\theta(t))$$



$$\frac{dy_{FM}(t)}{dt} = -\underbrace{A_c\left[2\pi f_c + 2\pi k_f m(t)\right]}_{\text{zarf}}\sin(\theta(t))$$

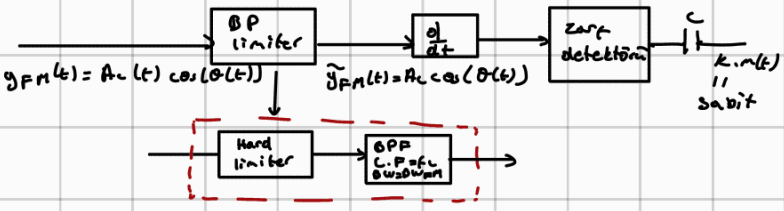
$\Rightarrow \forall t$ zarf > 0 olması gerekir

$\forall t \quad f_c + k_f m(t) > 0$

$\forall t \quad f_c > k_f \underbrace{\max|m(t)|}_{m_p}$

Δf

A_c sabit olmalıdır.



Pratik FM demodülatörleri

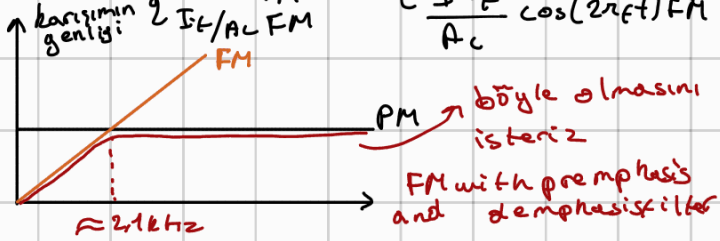
- Balanced modulators: slope detector / balanced discriminator / ratio detector
- Zero-crossed detector
- Phase Locked loop (PLL) (Düşük maliyetli ve yüksek performansı diğerlerine göre)

AM modülasyonunda karışım (interference)

$$r(t) = \underbrace{A_c\cos(2\pi f_c t)}_{\text{Unmodulated carrier}} + \underbrace{I\cos[2\pi(f_c+f)t]}_{\text{karışım}}$$

Mesajsız taşıyıcı

Demodülatör çıkışı: $y_d(t) = \begin{cases} \frac{I}{A_c}\sin(2\pi f t) & \text{PM} \\ \frac{I\cdot f}{A_c}\cos(2\pi f t) & \text{FM} \end{cases}$



Başlangıçta FM, sonra türevini alıp PM'e geçebilirsin. Bunu yapan filtreye ön vurgulama filtresi denir.

Additive White Gaussian Noise

AWGN: kanal gürültüsü

$$r(t) = y(t) + n(t)$$ ← AWGN